

PARTICULAR SPECIFICATION P2D

Appendix P2D - Contractor's Design and Build Obligations

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P2D APPENDIX P2D - CONTRACTOR'S DESIGN AND BUILD OBLIGATIONS

- (a) Requirements and Contractors Proposals
 - (i) The terms "Requirements" or "Employer Requirements" shall include (in order of precedence), the following:
 - 1) The Particular Conditions of Contract;
 - 2) The Standard Conditions of Contract;
 - 3) The Contractor's Design and Construction Requirements; including Codes of Practices, Standards, Regulations and Statutory Requirements
 - 4) The Particular Specifications;
 - 5) The General Specifications;
 - 6) The Drawings;
 - 7) Schedule of Prices;
 - 8) Annexes; and,
 - 9) Appendices.
 - (ii) However, where discrepancies occur the Contractor shall notify the S.O., who shall determine which is correct and advise the Contractor.
 - (iii) The term "Contractor's Proposals" shall include:
 - 1) The document called Contractor's Proposals or Contractor's Tender Submission (or any equivalent by any other name), and
 - 2) All drawings, specifications, schedules, calculations, manuals and other information submitted by the Contractor to S.O. in response to the Requirements, including any variations, amendments thereto.
- (b) Mandatory Requirements. Certain elements of the Works are mandatory to be provided for the Contract. The Contractor's Proposal must include and take into consideration the following elements and requirements:
 - (i) THP capacity and time and temperature performance
 - (ii) Sidestream N hydraulic capacity and removal performance

- (iii) Process Flow Diagram (PFD) and Piping and Instrumentation Diagrams (P&ID) are included in the Drawings. The Contractor shall note that the PFD and P&IDs show reference design with the minimum process requirements and functionality, including the minimum level of standby units/systems, and minimum level of instrumentation. The Contractor shall be responsible for the complete detailed design of the Works to ensure a safe and reliable fully functioning system.

P2D.2 General

- (a) The Drawings included in this Tender shall not be taken as limiting the responsibility of the Contractor for the study, investigation, detailed design and construction, testing and commissioning and performance proving of the Works.
- (b) The Contractor may change and modify the design for the Works to suit his proposal or design to meet the specified Requirements.
- (c) The Works to be executed under this Contract have been separated into various elements and are detailed in the relevant sections of the Particular Specifications, General Specifications and Schedules. A summary of the Scope of Works is also provided in Particular Specification P1 – Overview of the Works.
- (d) The Contractor shall be deemed to have included in the Contract Sum for all detailed design works and provide all Works which are not individually itemised in the Specifications but deemed necessary for the following aspects;
 - (i) To meet the specified performance requirements.
 - (ii) For complete system integration and safe, reliable functionality of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities including and all other ancillary plant and equipment, and
 - (iii) Include all provisions for safe access for operation and maintainability of all process systems, equipment, dosing, pumping and aeration systems, instrumentation, valves piping and tanks.
- (e) All detail designs, government authorities' submissions, supply, installation, construction, arrangement for authorities' inspections, testing and commissioning, including full performance testing and plant proving of the entire Scope of Works shall be provided by the Contractor.
- (f) Information on the arrangement for the conceptual process and indicative layout for the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities is provided in Volume 6. This conceptual design is provided for the process units and building layouts and are shown in Volume. The Contractor may adopt and further develop this conceptual design or modify or change this design to suit his proposal, and shall be responsible for all further investigation, study and detailed design of the Works.

- (g) Based on the design proposed by the Contractor together with the specified requirements, the Contractor shall validate and confirm the elevations, and dimensions for the proposed process units and buildings which are required to achieve the specified duty and to meet the performance guarantees required. The Contractor shall substantiate his design proposals by calculations. Once the S.O. has reviewed the Contractor's calculations in this respect and has agreed the final layout (including hydraulic profile where applicable) of the proposed Works, the Contractor shall carry out the detailed design of the all the process units and ancillary works.
- (h) For mechanical, electrical and instrumentation works, which does not have specific process guarantees, the Contractor shall select equipment of the type, quality, accuracy and number specified, and shall substantiate the selection by submitting detailed design calculations and drawings to the S.O. for the proposed duty of the equipment proposed.

P2D.3 Design Objectives

- (a) The Works under this Contract is to design, construct, commission and performance test the fully automated Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities to meet the Annex O Schedule of Performance Guarantees.
- (b) As a minimum, the design's objectives shall include the following aspects, which shall also be deemed to be performance criteria:
 - (i) **Consistent Performance** – Provide consistent treatment and treated sludge and used water quality performance of each of the treatment processes to achieve the specified treatment criteria;
 - (ii) **Plant Reliability** – Enable the reliability of the equipment, unit processes (unit operations) and plant as a whole to be demonstrated;
 - (iii) **Plant Redundancy** – Provide adequate redundancy of all unit processes and systems (i.e. capacity and/or number of units), to ensure availability of all equipment sufficient to operate, monitor and meet the plant performance requirements at all times;
 - (iv) **Plant Maintainability** – Provide safe access and isolation provisions such that all equipment, valves and instruments can be fully and safely maintained and calibrated without the need to shutdown the plant, process unit or subsystems;
 - (v) **Staff Training** – Enable operations staff to become familiar with the operations and maintenance aspects of such a plant;
 - (vi) **Cost Effectiveness** – Provide the operation costs (electricity, chemicals, etc.) of the system as a whole, and each unit process individually.

P2D.4 Design Features

- (a) The design features to be included in the design:
 - (i) For Digestion Pre-treatment Facility:
 - 1) Minimise energy use, both electricity and steam e.g. recovery of flash steam to minimise steam consumption
 - 2) Maximum heat recovery from Hydrolysed Sludge Cooling to produce hot water for the Greasy Waste Recovery Facilities.
 - 3) Minimum one (1) set of hot-standby master PLCs controlling Digestion Pre-Treatment facility, master PLCs shall communicate seamlessly peer-to-peer with other existing TWRP Plant PLCs via Plant Automation Network (PAN).
 - (ii) For Sidestream Nitrogen Removal Facility:
 - 1) Solids removal and equalisation , with the objective to improve the overall performance, availability and system uptime. Where possible, predictive monitoring and control system shall be included as part of the design element of the deammonification process to improve the operability.
 - 2) Design of the future provision for odour covers, odorous air extraction and odour control facility;
 - 3) All process tanks shall be partially covered as a minimum, with the aim to utilise the area above the tank for other equipment such as blowers. The contractor shall design the process tanks to ensure the open area coverage is adequate for natural ventilation. The partial covers shall be suitable to be odour control covers and the open areas shall be able to be fitted with odour covers without major modifications.
 - 4) Minimum one (1) set of hot-standby master PLCs controlling Sidestream Nitrogen Removal facility, master PLCs shall communicate seamlessly peer-to-peer with other existing TWRP Plant PLCs via Plant Automation Network (PAN)
- (b) The design shall minimise the overall footprint and space requirements, whilst providing all necessary clearances and operation and maintenance access provisions and accommodating all equipment required within the allocated area;
- (c) Where applicable, the design of the Works shall comply with all the authorities' requirements including making design changes to meet conditions if any, imposed by the authorities.

P2D.5 Design, Specifications and Other Information

- (a) The Contractor shall design and provide all necessary specifications for the Works in accordance with the plans and requirements in the Specifications including complying with authorities regulations.
- (b) All design details, plans, drawings, specifications, notes, annotations and information shall be provided by the Contractor in such sufficient format, detail, extent, size and scale and within such time as may reasonably be required to ensure effective execution of the Works and/or as otherwise required by the S.O. All dimensions shall be in metric units and all drawings shall be produced to adequate scale.
- (c) The Contractor shall be responsible for detail design of the Works to the extent and in accordance with the requirements of the Specifications. The Contractor may elect to configure his design to follow as closely as possible the layout, and building layouts shown on the Drawings prepared by the S.O., although these Drawings shall not be taken as limiting the responsibility of the Contractor.
- (d) The Contractor shall hold design workshops with the S.O. to confirm the design criteria, P&ID's, and operational and control philosophies. The process design and workshop discussions shall be documented in a design report. The design report shall include drawings for the proposed facilities. The Contractor shall prepare the necessary design report, P&ID's, operational and control descriptions and functional specifications, for the S.O.'s review and approval after the design workshops. The Contractor shall submit detailed design drawings of all elements, for which the Contractor bears design responsibility, for the review of and approval by the S.O., prior to purchasing of equipment and / or materials and commencing fabrication or construction of that element.
- (e) The conceptual design show in Volume 6 is for reference and indicative only. The Contractor shall prepare his own proposal consistent with his own experience and with the particulars and guarantees given by him in the Schedules.
- (f) The dispositions of conceptual design and of associated systems may be varied from those shown in the Drawings to suit particular requirements of plant and civil works. The Contractor shall develop the proposal with his own experience, the requirements of the Specifications and to fit the Works within the site boundary limits.
- (g) The design shown on the Drawings shall not constitute any obligation on the Employer. Although the proposals shown on the Drawings are intended to be recommendations to the Contractor except as above, the Contractor shall be fully responsible for proposals put forward by him. His designs shall reflect his own experience in works of a similar nature and magnitude, subject to the requirements of the Specifications and to site limitations and topography.

- (h) The Contractor shall carry out such surveys, studies and analyses of any kind as are necessary for the purpose of his general and detailed designs and to verify the information shown on the Drawings and other data furnished by the Employer. The Employer shall not be liable for any claim by the Contractor on account of incompleteness or inaccuracy of the Drawings and data furnished by the Employer irrespective of when the Contractor shall discover such incompleteness or inaccuracy.
- (i) The Contractor shall design and provide all necessary specifications for the Works in accordance with the requirements in the Specifications including complying with authorities regulations.

P2D.6 Submissions for S.O.'s Approval

- (a) The Contractor shall submit all design documents including drawings and design calculations to the S.O. for review and approval. The review and approval of the designs does not in anyway, restrict nor relieve the Contractor of any responsibility for the Contractor's errors or omissions or departure from the Contractor's design and construction obligations, Statutory Obligation or other requirements of this Contract.
- (b) The Contractor shall not make any material deviation, alteration, addition to or omission from the approved design without the prior written approval of the S.O.

P2D.7 Payment of Fees, Costs, etc.

- (a) All fees, costs and expenses incurred by the Contractor in carrying out the design and making submissions including re-submissions to relevant authorities for the Works whether direct, indirect and/or incidental
- (b) Where an Accredited Checker or Registered Inspector is required for the Works, they shall be engaged by the Employer. The Contractor shall be responsible for notifying the S.O. and Employer of the time schedule required for these services and allow ample time for the engagement of these services. The Contractor shall be responsible for the preparation of the structural design calculations, reports and drawings necessary for the Accredited Checker and Registered Inspector as required by the Building Control Act for the Works.
- (c) All fees or charges arising from application to authorities submission charges, processing fees, utility connection fees etc. All fees arising from application to authorities for information, searches are to be paid by the Contractor and deemed to be included in the Contract Sum.

P2D.8 Contractor's Design Responsibility

- (a) The Contractor shall be responsible for the complete detailed design, including all workshop drawings for the construction of the whole Works and obtain all relevant authorities' approvals (if required). The design shall be fully detailed and constructed in accordance with the specified requirements, complying with relevant authorities regulations and to the satisfaction of the S.O. and shall be undertaken with the skill and care of competent and registered professionals.
- (b) The Contractor warrants, absolutely and independent of fault, that the Contractor's final design meets the Requirements and that where there are any inadequacy, insufficiency, impracticality or unsuitability in or of the Requirements or any part thereof, the Contractor's design has taken into account, addressed or otherwise rectified such inadequacy, insufficiency, impracticality or unsuitability.
- (c) The Contractor shall indemnify the Employer against any damage, expense, liability, loss or claim which the Employer might incur, sustain or be subject to arising from any breach of the Contractor's design and construction responsibility and/or warranty set out in this Clause and under common law.
- (d) The Contractor further accepts full responsibility for his final design and the design of his consultants and/or his subcontractors and warrants, absolutely and independent of fault, that the same meets the requirements;
 - (i) Notwithstanding that such design may be or may have been prepared, developed or issued by any of the Contractor's consultant and/or his subcontractors or caused to be prepared, developed or issued by others;
 - (ii) Notwithstanding any warranties, guarantees and/or indemnities that may be or may have been submitted by any other person; and
 - (iii) Notwithstanding that the S.O. may have approved the same.
- (e) The Contractor shall be deemed to have reviewed all of the Contract Documents and shall be deemed to have satisfied himself that the drawn and written information provided in the Contract Documents are sufficient and adequate to enable him to prepare and complete the detail design and construction of the Works and bring the entire Works to full completion in accordance with the Contract Documents, complying with local authorities' regulations and to the satisfaction of the S.O.
- (f) The Contractor shall be responsible for the preparation and submission to the S.O. and relevant authorities of all design drawings, including fully detailed shop drawings, engineering data, test reports and structural calculations, including any modifications to the design.

- (g) All design including calculations and drawings shall be duly endorsed in accordance with the Building Control Act or other relevant local legislation by the Contractor's Professional Engineers of the appropriate discipline. The Contractor shall provide and be responsible for the detailed design of the following works but not necessary limited to:
 - (i) Site layout and general arrangements including necessary soil investigation and survey works;
 - (ii) Foundation for all structures for the Works;
 - (iii) All civil structures for the Works including external works;
 - (iv) All works for process mechanical, electrical and building services including external civil works;
 - (v) Complete thermal hydrolysis process and associated ancillary equipment;
 - (vi) Complete sidestream nitrogen removal process and associated ancillary equipment;
 - (vii) Suitability of all materials and equipment for their service, application and duty;
 - (viii) All minor and incidental works necessary to complete the entire Works.

P2D.9 Changes to Design and Works

- (a) The Contractor shall be responsible for incorporating all changes arising as a result of compliance to the S.O. and all authorities' requirements anytime during the Contract Period.
- (b) All designs shall be carried out according to the specified Standards and Codes of Practice (including Euro Codes) and shall comply with all Building and Safety Regulations including the Building Control Act. The Contractor shall be liable for any defects/damages arising out of non-conformity and all necessary remedial works shall be carried out at the Contractor's own cost and time.
- (c) The Contractor shall design and provide all necessary specifications for the design and construct of the Works in accordance with the requirements of the Contract with all elements suitable for the intended purposes in all respects. All design details, plans, drawings, specifications, notes, annotations and information shall be provided by the Contractor in sufficient form, detail, extent, size and scale to ensure effective execution of the Works and/or as otherwise required by the S.O.
- (d) The Contractor shall take into account in his design any constraints or effects imposed by the existing works and services in the surrounding areas and works of Other contractors.

- (e) The S.O. reserves the right to direct the Contractor's Qualified Person(s), Professional Engineers (PE) and Registered Architect (RegA) to amend the design calculations, design parameters and assumptions made. Should, in the opinion of the S.O., the design be inadequate, the Contractor's QP, PE and RegA shall be required to re-analyse and re-submit to the acceptance of the S.O. and the approval of all relevant government authorities. The acceptance by the S.O. of the design by the Contractor's QP, PE and RegA shall not in any way absolve the Contractor and his consultants of their responsibilities and obligations with regard to the entire design.
- (f) Approval by the S.O. and local authorities of the Contractor's proposals or of any subsequent revision thereof by the Contractor shall in no way relieve the Contractor of his entire responsibility for the adequacy and practicality of his proposals. The cost of all necessary temporary or permanent remedial works ordered by the S.O. or local authorities, as a result of any inadequacy discovered at any time:
 - (i) in the Contractor's tender proposal or
 - (ii) in the revised proposals, or
 - (iii) due to the Contractor's failure or inability to provide and complete the Works shall be the entire responsibility of the Contractor and shall be borne by him irrespective of whether such remedial work is performed by the Contractor himself or by some party authorised by the S.O.
- (g) The Contractor shall ensure that all design works are carried out expeditiously to secure early approval by the S.O. Thereafter, the Contractor shall proceed to secure the timely approval from all relevant authorities directly or through the S.O.. A detailed works programme indicating time frames required for the submissions, approvals, inspections and hand-over to the relevant authorities shall be submitted to the S.O. for approval. The Contractor's attention is drawn to the requirement that the Works must obtain Temporary Occupation Permit on or before the successful phase or part completion for Plant Proving Period.
- (h) The Contractor shall ascertain the location of all existing services and utilities that might interfere with or be endangered by the construction of the Works. The Contractor shall be deemed to have made due allowance for relocation of any existing services which may hinder the construction of the works in the Contract Sum.
- (i) Where modifications, amendments, revisions or alterations are required to be made to the design and/or to the Works at any time on account of site, safety, construction, statutory or any other constraints or reasons, the Contractor shall advise and notify the S.O. that the same is required and provided that written consent is given by the S.O., the Contractor shall provide, modify, amend, revise and/or alter all design details, and drawings, specifications, notes, annotations and information as may be necessary and shall carry out and complete the same.

- (j) The Contractor shall not execute any such modifications, amendments, revisions or alterations without the express and written consent of the S.O. In the event that modification of the design is necessary in order that the design meets the specified requirements then the Contractor shall not be entitled to any claims from the Employer.
- (k) All costs incurred by the Contractor as a result of design changes or additions arising from design reviews or any other design or operation and maintenance related workshop and required for a safe and fully functional, operable and maintainable plant, subsystems and all its associated provisions, whether direct, indirect and/or incidental shall be borne by the Contractor and shall be deemed to be included in the Contract Sum.

P2D.10 General Submission Requirement

- (a) The Contractor shall adopt and use for the whole duration of the Works the same document management system that is nominated by the S.O. for communication of project information, submission of project documentation for information, review, approval, etc. between the S.O. and the Contractor as described in Appendix P2C – Employer Information Requirements.
- (b) Hand-drawn drawings are not acceptable. All working drawings shall be prepared in a format approved by the S.O. Both electronic and hardcopy drawings should be submitted to the S.O. for each submission.
- (c) Design and materials that require the S.O.'s approval shall be submitted in advance of construction. For items that need time to source or design, adequate time shall be given to the S.O. to consider any proposed alternatives, as necessary.
- (d) All materials and workmanship shall comply with the Specifications as described in the Contract Documents. For items of work and materials where there are no specifications, the Contractor shall submit the relevant details to the S.O. for his approval. Details of any other matters which the Contractor knows or has reason to believe will materially affect the execution of the Works should also be given, and the cost shall be deemed to have been allowed for in his rates.

P2D.11 Detailed Design Calculations

- (a) The Contractor shall submit design calculations. All design calculations shall be in English, and clearly and systematically presented on A4 size papers (210 x 297 mm) weighing not less than 80 g/m². All calculation sheets shall be systematically filed in reasonably durable files to enable easy taking out or adding in of any calculation sheet. All files shall be numbered in order and with the title of the project, contract description, the names of the Contractor and his consultant(s) clearly stated on the front cover. The format of calculations shall follow that specified in the Building Control Act. All calculations shall be made available to the S.O. and provided as required by the S.O..

- (b) All analysis and design assumptions, design principles, loading conditions, computer programs used, references to Codes of Practice, technical literature etc. shall be clearly stated in the calculations. Where references to any technical literature are made, the relevant parts of the literature shall be included in the submission.
- (c) All calculations shall be carried out in S.I. Units as specified in BS 3763. The units of length shall be in metres (m) or millimetres (mm), force in kiloNewtons (kN) and stress in N/mm² or kN/m² (kPa).
- (d) All calculations including any amended or additional calculations shall be fully endorsed by the Contractor's PE and RegA.

P2D.12 Computer Programs

- (a) The Contractor shall submit details of any computer program that is used in the analysis and design. The details to be submitted shall include the following:
 - (i) The basis of using the programs.
 - (ii) The details of the programs such as the theoretical background, its capability, assumptions, methods used, input/output details etc.
- (b) The Contractor shall make available all computer programs to the S.O. for checking if this is deemed necessary by the S.O. at no extra cost.

P2D.13 Obtaining Approvals

- (a) The Contractor shall appoint a Registered Architect to co-ordinate and obtain any approvals required from the various relevant authorities through the S.O.. These may include but are not limited to, Building and Construction Authority (BCA), Ministry of Home Affairs (MHA), Singapore Civil Defence Force (SCDF), Fire Safety and Shelter Department (FSSD), PowerGas, City Gas, PowerGrid, Power Supply, Public Utilities Board (PUB), Info-Communications Development Authority (IDA), Ministry of the Environment (ENV), National Environment Agency, JTC Corporation (JTC), Urban Redevelopment Authority (URA), Housing and Development Board (HDB), Land Transport Authority (LTA), Ministry of Manpower (MOM), Defence Science & Technology Agency (DSTA), Civil and Aviation Authority of Singapore (CAAS), Energy Markets Authority (EMA), etc.
- (b) The Contractor shall ensure the design and build works stay on track by lining up permits and expediting the permit process for all parties. The Contractor shall be responsible for obtaining regulatory permits required for the Works. The Contractor shall ensure permit applications contain every required details in order to avoid delays in the review process. The Contractor shall coordinate closely with the S.O..

- (c) The Contractor shall appoint Qualified Persons. The Contractor's Qualified Persons shall submit directly all necessary forms required for the approvals and comply with all the authorities' requirements. Forms should be obtained from the respective authorities. The Contractor shall also submit the names and CVs of the Qualified Persons (QPs) proposed for the design and construct elements of the project. The proposed QPs shall be registered in compliance with the Building Control Act. The Contractor's QPs shall be responsible for the following direct authorities submission:
 - (i) BCA ST; and
 - (ii) FSSD M&E, ACMV and FP.
- (d) The Contractor's Registered Architect shall submit through the S.O. all necessary forms required for the approvals and comply with all the authorities' requirements. Forms should be obtained from the respective authorities. Except those listed in the **Error! Reference source not found.**(c), all authorities submissions, including providing the certificates from relevant statutory authorities or technical agencies for Temporary Occupation Permit (TOP) and Certificate of Statutory Completion (CSC) will be submitted through the S.O..
- (e) Except those listed in the **Error! Reference source not found.**(c), the Contractor shall be responsible for obtaining all authority approvals, permits and inspection clearances through the S.O.. The Contractor shall engage Professional Engineers of the appropriate discipline to carry out consultations with relevant technical departments, prepare all necessary drawings, and make all necessary submissions to the relevant authorities for the Works. The Contractor shall also be responsible for the application to obtain the Temporary Occupation Permit and Certificate of Statutory Completion through the S.O..
- (f) The Contractor shall also be responsible for detailed documentation as may be required for submission or resubmission to all and any of the various authorities for the purpose of obtaining approvals and permits on behalf of the Board.
- (g) The Contractor's programme shall consider the duration required for making consultations, applying waivers, obtaining all necessary clearance and approvals, directly or through the S.O., in order to achieve the key project milestones. Under any circumstances the Contractor shall not claim extension of time due to any delays resulting from obtaining approvals, directly or through the S.O., from the relevant authorities.
- (h) Except those listed in the **Error! Reference source not found.**(c), the Contractor shall be responsible for making consultations, obtaining all necessary clearances and approvals including Temporary Occupation Permit (TOP) and Certificate of Statutory Completion (CSC) from all relevant authorities including URA, BCA, FSSD, NEA, LTA, NParks, SPPG etc., necessary to implement, complete and handover the completed Works through the S.O..

P2D.14 Contractor's Design and Drawings

- (a) The Contractor shall ensure that the whole of his detailed drawings are satisfactorily designed to fully comply with the Specifications and are suitable for their purpose. The drawings shall be adequately detailed to indicate all fixing and support details and all other data and information as required or directed by the S.O.
- (b) The Contractor shall note that all drawings shall be produced in Building Information Modelling (BIM) format in accordance with Appendix P2C – Employer Information Requirements. This includes design drawings, drawings for approval, authorities' submission, Shop Drawings and 'As-Installed' or 'As-Constructed' Drawings.
- (c) The Contractor shall utilize BIM modelling software to carry out the design for the Works. The Contractor shall provide interactive 3D model review meetings for Employer's regular review of geospatial arrangement of the Works. Where required by the Employer, the Contractor shall provide 3D outputs including renderings and video fly/walk-through for presentations.
- (d) The Contractor shall submit electronic files representing the 3D model and extractions from the model. The model shall be used to prepare visualizations of the Works as required to convey the progress of and compliance of the design with the Employer's requirements.
- (e) The Contractor shall model the works with sufficient details to coordinate and facilitate interference checking of internal and external services for pipework, yard piping and Mechanical, Electrical, Instrumentation, Control and Automation (MEICA) services and also with civil structures as specified in Appendix P2C – Employer Information Requirements.
- (f) No work shall commence nor any equipment or material shall be fabricated, assembled, purchased, delivered or shipped until the drawings and specifications or other data are approved by the S.O., unless otherwise directed. The Contractor is responsible for seeing that only 'Approved' copies of drawings endorsed by the S.O. are allowed on the job.
- (g) The Contractor shall resubmit the drawings returned for amendments by the S.O. and carry out the necessarily amendments arising from comments made by the S.O. within fourteen (14) days. The Contractor shall be responsible for delays caused by rejections of inadequate or incorrect modifications drawings. He shall have no claim for additional cost and/or time if he fails to adhere to the resubmission schedule.
- (h) The Contractor shall keep adequate records of submission and approvals and an accurate up-to-date record file shall be kept in his site office at all times.
- (i) Approval of the Contractor's submission of drawings does not constitute a complete check but indicates only that the general design, method of construction and detailing are satisfactory. Approval by the S.O. does not mean that the Contractor is permitted to deviate from the contract requirements and does not relieve the Contractor of the responsibility for errors in dimensions, details, size of members, etc., or for co-ordinating installation and/or construction with actual conditions of the Works.

P2D.15 Format of Contractor's 3D/2D Drawings

- (a) All drawings prepared by the Contractor shall be drawn in a neat, clear and concise manner so that they can be correctly and expeditiously interpreted.
- (b) All BIM drawings shall comply with CORENET's BIM e-submission guidelines for Architectural, Structural and Mechanical, Electrical and Plumbing (MEP) disciplines respectively.
- (c) All BIM drawings shall be submitted in AECOSIM, REVIT or other approved file format compatible with the Employer's respective departments' Computer-Aided-Drafting System.
- (d) The format for the layout of the drawings with the prescribed margins and locations of information panels shall conform to A1-size drawings.
- (e) All abbreviations, notations and symbols used shall be standardised and clearly stated at the beginning of the set of drawings in accordance with format approved by the S.O., and consistently used throughout the drawings.
- (f) All drawings shall be accurate, detailed in S.I. Units (with lengths in metres (m) or millimetres (mm) and drawn to the following scales:-
 - (i) All layout plans - Not smaller than 1:200. However the Contractor is required to enlarge such plans to 1:100 upon award of Contract for construction purposes.
 - (ii) General arrangement such as sectional, elevations, details plan - 1:100 or 1:50
 - (iii) Details of structural elements such as slabs, beams, columns, etc. - 1:20
 - (iv) Architectural construction details - 1:20
 - (v) Blow-up details - Not smaller than 1:5
- (g) The scales used shall be clearly indicated in the title block of each drawing and preferably under the title of each view.

- (h) All drawings submitted shall be produced on papers weighing not less than 80 g/m². Drawings for each discipline shall be bound separately. All drawings for one discipline shall be neatly and securely bound together with the title of the project, the names of the Contractor and his consultant(s) clearly printed on the front cover.
- (i) In addition, the Contractor shall submit a complete set of As-Built Drawings in accordance with the provisions of the Contract. All As-Constructed Drawings shall be endorsed by the Contractor's PE and RegA.

P2D.16 Standards and Codes of Practice for Design

- (a) The Works shall be designed according to the latest editions of all relevant Standards and Codes of Practice including the latest Euro Codes for civil and structural works.
- (b) Additional Codes include the following:
 - (i) Electrical Works

Electricity Act (Cap. 89A)

The Electricity (Electrical Installations) Regulations
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- (ii) Mechanical Works

Code of Practice for Fire Precautions in Building

Handbook on energy conservation in Buildings and Building services
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Code of Practice on Environmental health
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and any other relevant current Singapore Standards, BCA, British Standards, European Standards, and other Standards, Codes and/or Guidelines issued by relevant authorities.

P2D.17 Ground Conditions

- (a) The Contractor shall be responsible for obtaining all supplementary information on the nature of the site and soil conditions (topographical survey soil investigation including existing services) as required for the execution of the Works. The Contractor shall allow for such works in his Contract Sum.

P2D.18 Ownership of Documents and Copyrights

- (a) The proprietary right in respect of Drawings, Specifications, design and other information prepared by the Contractor shall remain the property of the Contractor. However, should the Contractor's appointment be terminated the Employer shall be entitled to the exclusive use of all Drawings, Specifications, design and other information provided by the Contractor for the carrying out and completion of the Works and the Contractor shall not be entitled and shall be deemed to have waived all rights and claims for any payment or reimbursement for such use by the Employer.

P2D.19 Custody and Supply of Documents

- (a) Before the commencement of the Defects Liability Period the Contractor shall without further charge, supply for the retention and use of the S.O. such drawings and information showing or describing the Works as built, and concerning the operation and maintenance of the Works.
- (b) Before the commencement of the Defects Liability Period the Contractor shall without further charge, supply for the retention and use of the Employer all the required Asset Management System (AMS) information, including actual asset values in full accordance and compliance with Employer's AMS documentation requirements and software templates.
- (c) Neither party shall divulge or use except for the purposes of this Contract any of the documents, which the other party supplies to him.

P2D.20 Qualified Persons

- (a) The Contractor shall not replace such persons concurred by the S.O. as the qualified professionals/persons under these Conditions without the prior written concurrence of the S.O. to discontinue the same or to re-appoint and employ another. The S.O. shall have the authority to withdraw his concurrence of the qualified professionals/persons at any time. If such concurrence shall be withdrawn, the Contractor shall, after receiving notice in writing of such withdrawal, cease to employ or allow the qualified professionals/persons from carrying out any work or undertaking any duties in any capacity with respect to the Contract. The Contractor shall promptly replace them with qualified professionals/persons concurred by the S.O.
- (b) The Contractor shall be liable for all acts, omissions and default of such qualified professionals/persons.
- (c) Should the Contractor's appointment be terminated, the S.O. shall have the authority to continue or discontinue the same qualified persons or to re-appoint and employ another qualified persons. The Contractor shall promptly submit all necessary documents to facilitate the above.

P2D.21 Prices Inclusive

- (a) The Contractor shall be deemed to have included in the Schedule of Prices, all costs and expenses necessary for carrying out all that is necessary for completing all the Works in accordance with the Requirements. The Employer shall not entertain any claim whatsoever by the Contractor for any additional cost and expense.

P2D.22 Submissions of Designs

- (a) The Contractor shall comply with the Appendix P2E – Schedule of Submissions of drawings, calculations and other required documentation for the Employer's and S.O.'s review. Drawings, calculations and other required documentation to be submitted shall be in accordance with the requirements of this Specification and the accompanying General and Particular Specifications.
- (b) The floor plan to be submitted shall detail all equipment requirements for the plant. All process equipment for shall be included under this Contract.
- (c) The Contractor shall be responsible for the complete detailed design of the Works including its integration and functionality as a whole.
- (d) Prior to proceeding with the detailed design, the Contractor shall prepare the preliminary design, general arrangement drawings, P&IDs and functional specifications and submit to the S.O. for approval. The Contractor's Professional Engineer of the appropriate discipline shall endorse all such submissions to the S.O.
- (e) The Contractor shall submit documents and drawings necessary for all government authorities' submissions and full detailed design drawings for all elements of works under this Contract for the review and comment by the S.O, prior to purchasing equipment and/or materials and commencing fabrication or construction of the element. The Contractor's Professional Engineer of the appropriate discipline shall endorse the detailed design drawings and specification submissions to the S.O.
- (f) The Contractor shall prepare the necessary P&ID's, operational and control descriptions and functional specifications, for the S.O.'s review and approval, prior to undertaking detailed design, to ensure that the project meets the specified requirements. The Contractor shall submit detailed design drawings of all elements, for which the Contractor bears design responsibility, for the review of and approval by the S.O., prior to purchasing of equipment and / or materials and commencing fabrication or construction of that element.

P2D.23 Safety in Design (SID), Hazard and Operability Study (HAZOP) and Access Lifting Maintainability Operability (ALMOP) Workshops

- (a) During the detailed design phase of the works, the Contractor shall carry out Safety in Design (SID) review workshops and other key workshops that as a minimum shall include:
 - (i) Hazard and Operability (HAZOP) Studies and workshops;
 - (ii) Access Lifting Maintainability and Operability (ALMOP) Workshops;

- (iii) Safety in Design (SID) reviews and workshops;
 - (iv) Flood and Fire Risk Assessment for all areas and facilities;
 - (v) Functional Safety Review Workshop;
 - (vi) Failure Mode Effect and Criticality Analysis; and
 - (vii) Constructability Workshops.
- (b) The Contractor shall ensure that equipment and facilities are designed to be safe throughout their life cycle – from design through construction, operation and maintenance. An appropriate risk identification and mitigation approach should be adopted, which shall involve dedicated workshops to scrutinise the design against a prepared list of safety criteria and likelihood, consequence and mitigation strategies determined by the project stakeholders and plant operators.
- (c) This detailed design process shall include Safety in Design (SID) guidelines, processes and tools as appropriate. The Contractor's approach to SID shall typically be a risk identification and mitigation approach, following guidelines from the Board of Architects, Professional Engineers Board and Workplace Safety & Health.
- (d) The Contractor shall assign a qualified professional engineer to serve as the design for safety coordinator for this project. His responsibilities will be to ensure that the final designs are safe, functional and operable and maintainable. This is particularly important where there are significant construction or operational risks as a result of the site constraints.
- (e) The Contractor shall submit a schedule of SID Workshops to the S.O. and provide exact details and invites at least 4 weeks prior to each workshop such that all necessary stakeholders and Employer's staff can be made available to attend.
- (f) The Contractor shall arrange full HAZOP studies and HAZOP workshops for the final detailed design and equipment layout of the plant.
- (g) A HAZOP is a detailed systematic examination of a process and/or design to identify and assess the potential hazards or operability issues and the proposed methods of their control. It is undertaken by a group consisting of design, operations and maintenance personnel who have a direct involvement in the process. The HAZOP procedure is facilitated by a comprehensive checklist of guidewords which act as "prompts". It facilitates the recognition of hazards and potential operational problems which can be avoided or mitigated by changes in design or procedures.
- (h) The principal objectives of the HAZOP study are:
- (i) Systematically examine the proposed systems and equipment in order to identify the potential hazards to people, plant assets, operations, and the environment.
 - (ii) Assess the hazards and the proposed methods of control and decide on appropriate actions which will enable the risks to be reduced or eliminated.

- (iii) Assess the design and operations controls and their effects on the potential hazards.
- (iv) Document the proposed actions and assign individuals to be responsible for completing each action.
- (i) The HAZOP study/workshop shall be held at the Contractor's premises and shall cater for all interested parties. This called the HAZOP Team and shall include Employer's representatives (plant operators, maintenance personnel etc.), the S.O. and S.O's representatives, design team, members and consultants.
- (j) The Contractor shall submit a schedule of HAZOP Workshops to the S.O. and provide exact details and invites at least 4 weeks prior to each workshop such that all necessary stakeholders and Employer's staff can be made available to attend.
- (k) The Contractor shall engage an independent, external HAZOP facilitator for the duration of the HAZOP. All 30% design stage drawings (layouts, P&IDs etc.) and functional descriptions and operating philosophies are to be available for the HAZOP workshop and 30 copies of all documentation shall be delivered to the S.O. at least 2 weeks, prior to date of the HAZOP workshop. It is expected that the HAZOP study be carried out over at least 3 days.
- (l) An ALMOP study is a detailed systematic examination of a facility to identify and assess the potential hazards or operability issues associated with the operational and maintenance tasks that will need to be performed during the operational life of the facility. It is undertaken by a group consisting of design, operations and maintenance personnel who have a direct involvement in the process. The ALMOP procedure is facilitated by a comprehensive checklist of guidewords which act as "prompts". It facilitates the recognition of hazards and potential operational problems which can be avoided or mitigated by changes in design or procedures.
- (m) The principal objectives of the ALMOP study are:
 - (i) Systematically examine the proposed systems and equipment in order to identify the tasks required to be performed. The identification of the access, lifting and other facilities that will enable these tasks to be performed safely and to minimise any hazards
 - (ii) For each task the Contractor's design engineer will be required to: describe the design intent – how the proposed equipment relates to the task under consideration; and describe the design features in the current design that allow the task to be performed effectively and safely without compromising other functions.
 - (iii) Assess the safety and hazards, and the proposed design and provisions. Decide on appropriate actions which will enable the risks to be reduced or eliminated.
 - (iv) Document the proposed actions and assign individuals to be responsible for completing each action.

- (n) The ALMOP study/workshop shall be held at the Contractor's premises and shall cater for all interested parties. This called the ALMOP Team and shall include Employer's representatives (plant operators, maintenance personnel etc.), the S.O. and S.O's representatives, design team, members and consultants.
- (o) The Contractor shall submit a schedule of ALMOP Workshops to the S.O. and provide exact details and invites at least 4 weeks prior to each workshop such that all necessary stakeholders and Employer's staff can be made available to attend.
- (p) The Contractor shall engage an independent, external ALMOP facilitator for the duration of the ALMOP. All 80% design stage 3D models and the operating & maintenance task list are to be available for the ALMOP workshop and 30 copies of all documentation shall be delivered to the S.O. at least 2 weeks, prior to date of the ALMOP workshop. It is expected that the ALMOP study be carried out over at least 3 days.
- (q) Where modifications, amendments, revisions or alterations are required to be made to the design and/or to the Works at any time on account of site, safety, construction, statutory or any other constraints or reasons, the Contractor shall advise and notify the S.O. that the same is required and provided that written consent is given by the S.O., the Contractor shall provide, modify, amend, revise and/or alter all design details, and drawings, specifications, notes, annotations and information as may be necessary and shall carry out and complete the same.
- (r) All costs incurred by the Contractor as a result of design changes or additions arising from design reviews, design workshops, Safety in Design workshops, HAZOPs, ALMOPs or any other design or operation and maintenance related workshop and considered by the S.O to be required to fulfil the requirement for a safe and fully functional, operable and maintainable plant, its subsystems and all its associated provisions shall be borne by the Contractor and shall be deemed to be included in the Contract Sum.
- (s) The Contractor shall be responsible and deemed to have allowed time for coordination, interfacing and cooperation with other contractors and the Employer's personnel in the execution of the Works to achieve the programme requirements, sharing of work space and site areas where necessary.
- (t) The Contractor shall also be responsible and deemed to have allowed time for coordination and execution of all operational safety and maintenance activities associated with the process equipment and systems until the Completion of the Whole of the Works.

END OF SECTION